

Primary VPT Chain: ProducerUA -> ProZorro -> Retail

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely I(1)-like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

standardized_type=cottage_cheese

retailer=novus

promo_variant=promo_controlled

frequency=daily

Primary chain is strictly ProducerUA -> ProZorro -> Retail. Retail is estimated separately for silpo, novus, and silpo_novus. Combined rule: daily median of available silpo and novus standardized_type prices.

Table: PreTests

series	adf_level_p	kpss_level_p	adf_diff1_p	kpss_diff1_p	adf_diff2_p	kpss_diff2_p	integration_class	stability_flag	standardized_type	retailer	promo_variant	cointegration_p	pair_n_obs
producer	nan	nan	nan	nan	nan	nan	ambiguous	0.0	cottage_cheese	novus	promo_controlled	nan	nan
prozorro	1.1483227108295172e-10	0.01	4.589982408975803e-17	0.1	4.813752592554151e-10	0.1	I(1)	0.0	cottage_cheese	novus	promo_controlled	nan	nan
retail	nan	nan	nan	nan	nan	nan	ambiguous	0.0	cottage_cheese	novus	promo_controlled	nan	nan
producer_to_prozorro	nan	nan	nan	nan	nan	nan	pair	nan	cottage_cheese	novus	promo_controlled	nan	0.0
prozorro_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	cottage_cheese	novus	promo_controlled	nan	17.0
producer_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	cottage_cheese	novus	promo_controlled	nan	0.0

Table: ModelCoefficients

standardized_type	retailer	promo_variant	frequency	link	model_family	y_series	x_series	n_obs	sr_coef	lr_coef	ect_coef	ect_pvalue	asymmetry_short_p	asymmetry_long_p	vecm_rank	model_status	unreliable_flag	notes	primary_chain_doc
cottage_cheese	novus	promo_controlled	daily	none	none	n/a	n/a	203	nan	nan	nan	nan	nan	nan	nan	unavailable	1	No admissible model estim...	Primary chain is strictly...

Table: ResidualDiagnostics

model_family	link	y_series	x_series	ljungbox_p	arch_p	jb_p	unreliable_flag
none	none	n/a	n/a	nan	nan	nan	1

Table: SeriesUsed

date	producer	prozorro	retail	standardized_type	retailer	promo_variant	combined_rule
2025-04-14	nan	138.0	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-04-16	nan	118.8	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-04-18	nan	108.9	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-04-21	nan	120.0	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-04-22	nan	99.0	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-04-23	nan	107.08	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-04-24	nan	160.2	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-04-29	nan	156.06	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-04-30	nan	226.25	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-05-01	nan	82.5	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-05-02	nan	118.08000000000001	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-05-05	nan	101.29499999999999	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-05-06	nan	115.0	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-05-07	nan	258.2	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-05-08	nan	145.4	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-05-09	nan	101.0	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-05-11	nan	89.4	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-05-12	nan	168.0	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-05-13	nan	131.34	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-05-14	nan	75.0	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-05-15	nan	152.0	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-05-16	nan	145.2	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-05-19	nan	98.75	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-05-20	nan	159.0	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-05-21	nan	134.04	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-05-22	nan	139.98000000000002	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-05-23	nan	107.55	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-05-26	nan	168.6	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-05-27	nan	156.06	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-05-29	nan	150.5	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-05-30	nan	165.0	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-06-02	nan	135.0	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-06-03	nan	122.0	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-06-05	nan	142.0	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-06-06	nan	90.52000000000001	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-06-09	nan	60.0	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-06-10	nan	112.98	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-06-11	nan	93.0	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-06-12	nan	108.0	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...
2025-06-13	nan	71.5	nan	cottage_cheese	novus	promo_controlled	daily_median_of_available...

Table: LagProfile

	pair	lag	corr
prozorro_to_retail	1	-0.08256428191281034	
prozorro_to_retail	2	0.03971723294061452	
prozorro_to_retail	3	-0.28299691370846675	
prozorro_to_retail	4	0.26565078792035735	
prozorro_to_retail	5	0.1303041354472621	
prozorro_to_retail	6	-0.26925447699907323	
prozorro_to_retail	7	-0.17216709836907204	
prozorro_to_retail	8	0.12956079796648345	
prozorro_to_retail	9	0.11206095766065338	
prozorro_to_retail	10	0.2040978126322994	
prozorro_to_retail	11	0.05589447735426344	
prozorro_to_retail	12	0.1138962180039834	
prozorro_to_retail	13	-0.1824878791281883	
prozorro_to_retail	14	-0.3814313315470263	
prozorro_to_retail	15	-0.37383558079104423	
prozorro_to_retail	16	-0.02635756214685676	
prozorro_to_retail	17	-0.08611657567161285	
prozorro_to_retail	18	-0.023291506094282332	
prozorro_to_retail	19	0.32564562718987905	
prozorro_to_retail	20	0.4207617993239002	
prozorro_to_retail	21	-0.4536759346531177	
prozorro_to_retail	22	-0.5409233011486455	
prozorro_to_retail	23	-0.0802734474776812	
prozorro_to_retail	24	0.2944740100850855	
prozorro_to_retail	25	0.4074961775359445	
prozorro_to_retail	26	-0.1708554826531992	
prozorro_to_retail	27	-0.1981394710393736	
prozorro_to_retail	28	0.3893050688977193	
prozorro_to_retail	29	0.15315998800417385	
prozorro_to_retail	30	-0.016791942664708615	

Table: ModelEligibility

standardized_type	retailer	promo_variant	frequency	integration_producer	integration_prozorro	integration_retail	n_obs_producer	n_obs_prozorro	n_obs_retail	n_obs_pair_prod_prozorro	n_obs_pair_prozorro_retail	any_i2	eligibility_note
cottage_cheese	novus	promo_controlled	daily	ambiguous	I(1)	ambiguous	0	190	20		0	17	0 Low pair overlap in main ...

Table: NARDL_Multipliers

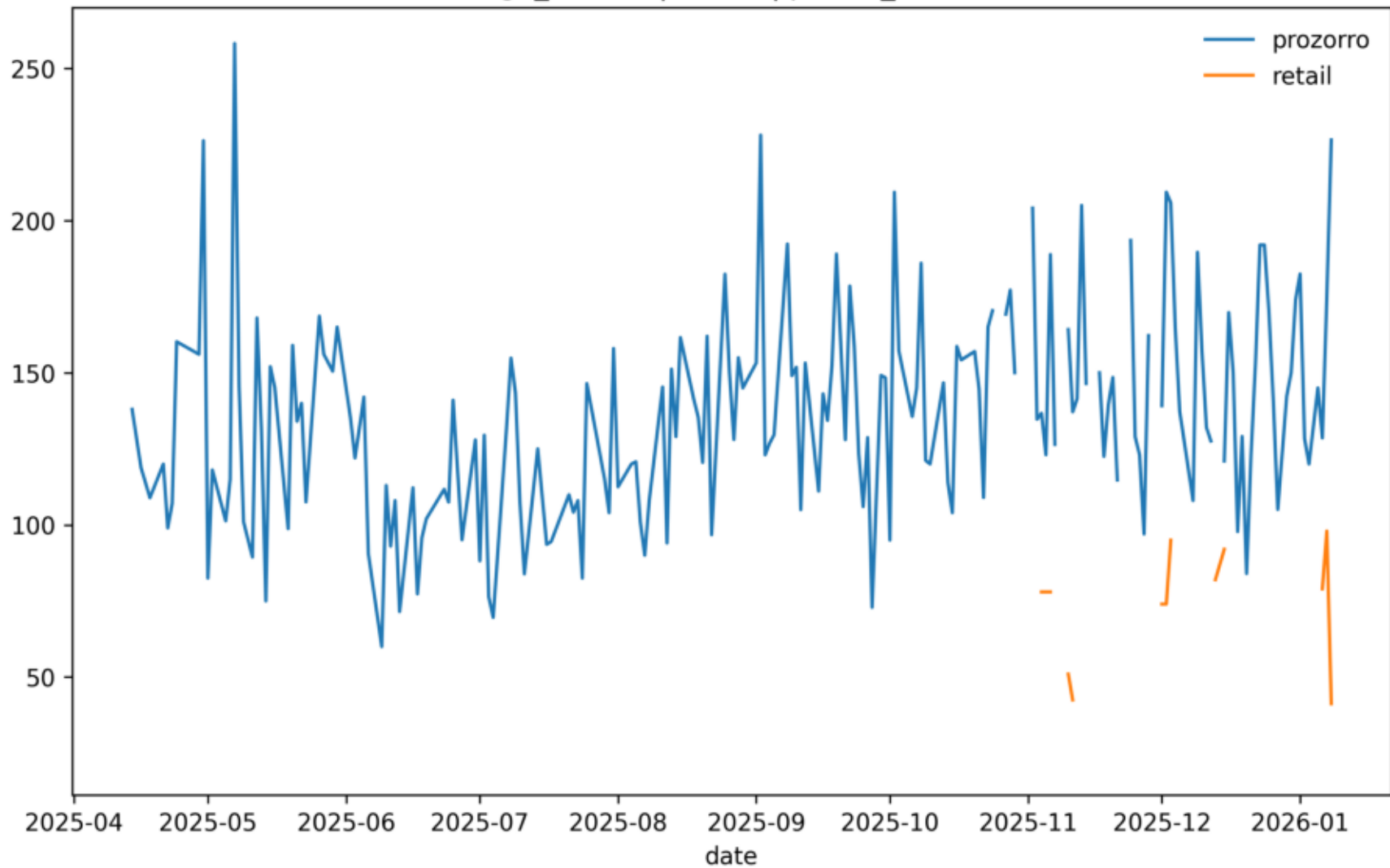
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Table: VECM_IRF

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time_series_promo_controlled.png

cottage_cheese | novus | promo_controlled



Lag-Correlation Profile | cottage_cheese | novus | promo_controlled

